

William Huanshan Chuang

Marana, AZ | williamchuang@arizona.edu | williamchuang.github.io | *U.S. national security-aligned R&D*

Profile

Founder, **Logarcheon Inc.** (secure AI & encrypted cognition). Independently researches and **volunteers** results to strengthen lawful U.S. mission readiness (OSINT/ISR, post-quantum cryptography, resilient autonomy). Research is conducted independently with **ITAR/EAR** awareness; sensitive methods are withheld and only **unclassified derivatives** are prepared for lawful dissemination to allied stakeholders.

Core Research Domains

- **Secure AI & Attention:** critical-entropy attention (CEAS), dynamic- β scaling, thermodynamic synchronization, interpretable control under adversarial stress.
- **Post-Quantum Cryptography:** MSIA; modular trace/zeta constructs; modular spectral indistinguishability; symbolic entropy.
- **Encrypted Cognition & ISR:** multimodal sensing (non-contact psychophysiology), adaptive signal fusion, GPS-/comms-denied autonomy.

Experience

Founder, Logarcheon Inc.

Jun 2025 – Present

United States (stealth R&D)

- Architect secure AI and encrypted cognitive systems: symmetry-governed attention (CEAS, dynamic- β) and interpretable control surfaces.
- Design post-quantum primitives (MSIA; modular spectral indistinguishability) and auditable cryptographic pipelines.
- Build OSINT/ISR-aligned prototypes for GPS-/comms-degraded environments; emphasize resilience and lawful guardrails.
- Establish workflows mindful of **ITAR/EAR** and export-control requirements; sensitive outputs withheld, unclassified derivatives prepared for appropriate dissemination.

Patents (Provisional & Pending Nonprovisional)

- **Critical Entropy Attention System (CEAS).** U.S. Provisional App. #63/813,617 Filed: May 29, 2025
- **Analytic Corridor Methods for Critical Synchronization, Memory Integration, and Optimization in Ising-like Neural Systems with Applications to Mind Upload, Modular Restoration, and AI Global Optimization.** U.S. Provisional App. #63/824,102 Filed: Jun 15, 2025
- **System and Method for Remote Polygraphy and Internal Voice Capture for Neural Training via Multimodal Electromagnetic Sensing.** U.S. Provisional App. #63/824,053 Filed: Jun 15, 2025
- **Neural Projection Unit with CEAS-Driven Dynamic Beta Scaling and Thermodynamic Synchronization.** U.S. Nonprovisional App. #19/232,791 Filed: Jun 9, 2025
- **Neural Projection Unit with CEAS-Based Dynamic Beta Scaling and Thermodynamic Synchronization.** U.S. Provisional App. #63/820,652 Filed: Jun 9, 2025
- **The Modular Symbolic Intelligence Architecture (MSIA).** U.S. Provisional App. #63/809,257 Filed: May 20, 2025
- **Hyperbolic Autoencoder and Transformer Framework for Real-Time Landscape Modification and Spacetime Adaptation.** U.S. Provisional App. #63/773,441 Filed: Mar 18, 2025

Education

University of Arizona — M.S., Mathematics

May 2025

Graduate-level coursework completed under Ph.D. classification; Graduate Teaching Assistant (College Algebra, Calculus I/II).

San Francisco State University — M.A., Mathematics

May 2022

Thesis: *The Hausdorff Dimension of Limit Sets of Well-Distributed Schottky Groups*.

University of San Francisco — B.S., Mathematics; Minor, Computer Science

Dec 2018

Affiliations & Service

- **Space Force Association (SFA)** — Member, 2025–present
- **Order of Malta** — Auxiliary Member (Arizona Area)
- **Knights of Columbus** — Fourth Degree (Patriotic Degree)

Technical Stack

Python (PyTorch, Lightning, NumPy, Pandas), C/C++, R, Mathematica, Git; formal methods (Lean 4); geometric & spectral methods; secure ML; OSINT workflows.

Legal & Ethical Note: Research is self-initiated and complies with U.S. law & export controls. Sensitive technical details remain undisclosed and will only be shared under appropriate clearance or other lawful mechanisms. Results are offered **voluntarily** to strengthen allied resilience and U.S. mission readiness.